25900250 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTG202 - FUNDAMENTALS OF MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Mention two proteins that are associated with bacterial nucleoid organization [U] / [CO1]
2. What is the function of DNA Polymerase ? [K] / [CO2]
3. What is the holoenzyme of RNA polymerase made up of? [A] / [CO3]
4. Name the two types of chromatin. [K] / [CO1]
5. Which enzyme catalyzes the addition of poly (A) tail to mRNA ? [K] / [CO3]
6. Recall the DNA form that appears in dehydrated conditions. [K] / [CO1]
7. Identify the role of the promoter in gene regulation. [U] / [CO3]
8. Mention two ways in which translation can be regulated in eukaryotes. [K] / [CO3]
9. Name the scientists who discovered the structure of DNA, and the year. [K] / [CO1]

10. In negative regulation, what happens when the repressor is bound to the operator? [U] / [CO3]

11. Name the strains of Streptococcus pneumoniae used in Griffith's experiment. [K] / [CO1]

12. Name the host plant commonly infected by TMV. [K] / [CO1]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. Differentiate between mRNA and tRNA in terms of structure and function. [U] / [CO1]

14. Explain photoreactivation as a direct DNA repair mechanism. [U] / [CO2]

15. Describe the role of reverse transcriptase in HIV replication. [U] / [CO3]

16. Why is DNA replication described as semi-conservative? [K] / [CO2]

17. Write a note on glycosylation of proteins and its biological significance. [U] / [CO3]

18. Differentiate between structural gene and regulatory gene. [U] / [CO1]

19. Why is lac operon considered an inducible operon? Relate to cAMP and CAP. [U] / [CO3]

[4x5 = 20]

Part C

Essay type questions

Answer any **1** questions

Each question carries **10** marks

20. With a neat labelled diagram explain the replication of DNA in eukaryotes. [U] / [CO2]

21. Describe in detail how translation occurs in prokaryotes. [K] / [CO3]

[10x1 = 10]